NBA Player Performance & Trade Value Predictor

DATA 300 Final Project Report

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Abstract

This project develops a machine learning pipeline for predicting NBA player performance and estimating player trade value. Using historical NBA data scraped from Basketball-Reference, the project combines data collection, feature engineering, regression modeling, clustering, and a custom trade value scoring system to evaluate player quality and long-term value. The system attempts to answer a practical sports analytics question: how can teams identify valuable players before the rest of the league?

The project is divided into five major notebooks:

1. Data Collection
2. Feature Engineering
3. Regression Modeling
4. Player Archetype Clustering
5. Trade Value Score & Leaderboard

Together, these notebooks create a complete end-to-end sports analytics workflow.

1. Data Collection

The first notebook focuses on collecting NBA player statistics from Basketball-Reference. Historical data was scraped for multiple NBA seasons using Python.

The project collected both:

- Per-game statistics
- Advanced analytics statistics

Examples of collected variables include:

- Points per game (PTS)
- Assists (AST)
- Rebounds (TRB)
- Player Efficiency Rating (PER)
- Box Plus/Minus (BPM)
- Win Shares (WS)
- True Shooting Percentage (TS%)
- Value Over Replacement Player (VORP)

The notebook uses helper scraping functions to automatically access Basketball-Reference URLs and extract tables into pandas DataFrames.

Once collected, datasets were cleaned and merged into a combined player dataset. Duplicate rows and formatting inconsistencies were removed to ensure data quality.

The final output of this stage was a cleaned multi-season NBA player dataset used throughout the remainder of the project.

2. Feature Engineering

The second notebook focused on transforming raw player statistics into predictive features suitable for machine learning.

Two major feature categories were created:

Lag Features

Lag variables represent a player's previous-season statistics. For example:

- PER_lag1
- BPM_lag1
- WS_lag1

These features capture the idea that recent past performance is often predictive of future performance.

Rolling Features

Rolling features use multi-year windows to capture long-term trends.

Examples include:

- 3-year rolling averages
- Rolling standard deviations
- Trend indicators

These features help identify:

- Consistency
- Improvement trajectories
- Decline patterns
- Injury-related volatility

Feature engineering was one of the most important stages of the project because high-quality features significantly improve predictive model performance.

The notebook ultimately produced a processed feature dataset used in regression modeling.

3. Regression Modeling

The third notebook built predictive machine learning models for forecasting future NBA player performance.

The project evaluated several machine learning approaches, including:

- Ridge Regression
- Random Forest
- Gradient Boosting Models
- K-Nearest Neighbors (KNN)

The target variables included advanced player metrics such as:

- Future PER
- Future Win Shares
- Future Points Per Game

Cross Validation

The project used 5-fold cross-validation to evaluate model performance and reduce overfitting risk.

Cross-validation allowed the project to compare:

- Lag-only feature sets
- Rolling feature sets
- Different machine learning models

This helped determine which features and algorithms generalized best to unseen player data.

Scaling and Pipelines

Pipelines with StandardScaler were used for models sensitive to scale, especially:

- Ridge Regression
- KNN

Tree-based models such as Random Forests and Gradient Boosting were less sensitive to scaling.

Results

The regression models successfully generated strong predictive performance across multiple target variables. For future PER prediction, the Ridge Regression model achieved a test R^2 of 0.828 with an RMSE of 1.713. Future points per game predictions performed even better, with the Gradient Boosting model producing a test R^2 of 0.850 and RMSE of 2.294. The Win Shares prediction model achieved a test R^2 of 0.734 with an RMSE of 1.137.

Feature importance analysis also showed that previous-season performance was highly predictive of future production. For example, PER_lag1 produced a correlation of 0.809 with future PER, while PTS_lag1 produced a correlation of 0.830 with future scoring output. Rolling three-year averages also remained highly predictive, with PER_roll3_mean and PTS_roll3_mean producing correlations above 0.79 and 0.82 respectively.

These predictions later became key inputs in the final trade value scoring system.

The regression stage demonstrated how machine learning can be used to forecast player development and future contribution.

4. Player Archetype Clustering

The fourth notebook used unsupervised machine learning to identify NBA player archetypes.

Rather than grouping players by simple counting statistics, the clustering model focused on role-defining efficiency and playstyle metrics.

Examples of clustering variables included:

- TS%
- BPM
- VORP
- Assist Percentage

- Usage Rate
- Defensive metrics

K-Means Clustering

The clustering analysis was performed on 6,326 NBA player-seasons. K-Means clustering was used to divide players into groups with similar statistical profiles.

The project evaluated the optimal number of clusters using:

- Elbow Method
- Silhouette Score

These techniques helped identify the most meaningful cluster structure.

Interpretation of Clusters

The project selected six final archetype clusters using the Elbow Method and silhouette score evaluation. The final clustering solution achieved a silhouette score of 0.1755 and an Adjusted Rand Index (ARI) of 0.4111 when compared against hierarchical clustering methods.

The resulting clusters represented different player archetypes, such as:

- Primary scorers
- Playmakers
- Defensive specialists
- Role players
- Efficient big men

The final archetype counts included:

- Defensive Anchors: 1,960 players
- Stretch Bigs: 1,289 players
- Utility Bench Players: 1,085 players
- Playmaking Guards: 842 players
- Elite Scorers: 685 players
- 3-and-D Wings: 465 players

Clustering added an important contextual layer to the project because players should not only be evaluated by raw production, but also by the type of role they play.

5. Trade Value Score & Leaderboard

The final notebook combined the outputs of previous notebooks into a custom Trade Value Score.

The goal was to estimate a player's overall trade value by combining:

- Current production
- Predicted future production
- Efficiency
- Age
- Positional/archetype context

Trade Value Components

The score used multiple normalized components, including:

1. Projected output score
2. Efficiency score
3. Age/upside score
4. Archetype value adjustments

Predicted statistics from the regression notebook were merged with clustering outputs to produce a final player ranking system.

The project then generated a trade value leaderboard ranking 2,367 NBA players based on estimated long-term value.

The final rankings also produced realistic valuation trends that aligned with modern NBA roster construction principles. For example, 38 of the 50 most overvalued players in the dataset were age 32 or older, suggesting the model appropriately penalized aging curves and declining long-term upside. Additionally, 50 of the top 50 highest-value players were identified as being on rookie-scale or minimum-level contracts, reflecting the importance of young, cost-controlled talent in NBA front-office decision making.

This stage demonstrated how predictive analytics and unsupervised learning can be combined into a practical front-office decision-making framework.

Methodology Summary

The project follows a complete data science pipeline:

1. Collect historical NBA data
2. Clean and merge datasets
3. Engineer predictive features
4. Train and evaluate machine learning models
5. Cluster player archetypes
6. Build a composite trade value metric
7. Rank players based on projected value

This workflow reflects modern sports analytics practices used by NBA organizations.

Strengths of the Project

Several aspects of the project strengthen its analytical quality:

- Multi-stage machine learning pipeline
- Combination of supervised and unsupervised learning
- Cross-validation for robust model evaluation
- Use of advanced NBA analytics metrics
- Practical real-world application

The project also demonstrates strong integration between feature engineering, predictive modeling, and interpretation.

Limitations

Despite strong results, the project has several limitations.

First, player performance is influenced by factors difficult to quantify statistically, including:

- Injuries
- Team chemistry
- Coaching systems
- Psychological factors
- Contract situations

Second, the model relies heavily on historical performance, meaning sudden breakout seasons or unexpected declines may be difficult to predict.

Third, trade value in real NBA decision-making also depends on salary cap constraints and organizational strategy, which were outside the scope of this project.

We also had trouble with the output table of trade value scores. It kept outputting multiple rows for the same player because if they got traded mid season.

Future Improvements

Several future improvements could strengthen the project:

- Incorporating salary and contract information
- Adding injury history variables
- Using deep learning sequence models
- Including playoff performance metrics
- Expanding clustering with advanced tracking statistics
- Building an interactive dashboard for trade simulations

These additions could make the system even more useful for practical front-office analysis.

Conclusion

This project demonstrates how modern machine learning techniques can be applied to professional sports analytics. By combining data scraping, feature engineering, predictive modeling, clustering, and scoring systems, the project creates a comprehensive framework for evaluating NBA player trade value.

The final system goes beyond traditional box score analysis by integrating predictive analytics and player archetype modeling into a single decision-making tool. The project highlights the growing role of data science in professional sports and shows how machine learning can support roster construction and player evaluation.

Overall, the project successfully builds a scalable and interpretable NBA trade value prediction pipeline with meaningful real-world applications.